HIRBOD JABBARNEZHAD

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Summary

Computer Science senior focused on AI integration and applied AI development, with hands-on experience building AI-powered applications, integrating external AI APIs, developing backend services, and deploying machine learning systems. Skilled in Python, FastAPI, REST APIs, PyTorch, Firebase, and Google Cloud.

Skills

- **Languages:** Python, Java, C++
- **Technologies:** FastAPI, REST APIs, Firebase, Google Cloud, PyTorch
- **Tools:** Jira, Git, GitHub, Figma, Open AI Codex
- **Concepts:** Data Structures & Algorithms, OOP, Agile

Experience

Satellite Change Detector 03/2026 - 06/2026
ML Computer Vision Project
- Built a Siamese U-Net from scratch to detect surface changes between satellite image pairs, achieving 0.70 IoU and 0.79 F1 on the LEVIR-CD benchmark
- Trained on a combined dataset of 1,083 high-resolution Google Earth image pairs (LEVIR-CD + LEVIR-CD+) using PyTorch with ResNet-34 encoder and BCE + Dice loss
- Deployed as a live interactive web app on Hugging Face Spaces with real-time heatmap overlay, binary change mask, and per-pixel probability visualization
- Implemented Test Time Augmentation (TTA) at inference to improve prediction robustness without additional training

GoneVishing – AI Scam Call Defense System 09/2025 - 04/2026
Software Engineering Project | Northridge, California
- Integrated Twilio and ElevenLabs APIs to enable a real-time AI voice agent that engages with scam callers
- Built backend services using FastAPI to handle call routing, API integration, and real time processing
- Designed call-handling logic and implemented storage of call metadata and transcripts using Firebase and Google Cloud

Schedule Builder — Web Application | **Live Demo:** https://esl-schedule-builder.netlify.app/
- Developed a web app that generates customized schedules based on user preferences and constraints.
- Built scheduling logic to detect conflicts and generate valid schedule combinations.
- Designed an intuitive interface for comparing and selecting schedules.
- Deployed and tested the application with iterative usability improvements.

Education

Bachelor of Science: Computer Science Expected in 12/2026
California State University, Northridge | Northridge, CA
- Dean's List
- Member, Tau Beta Pi Engineering Honor Society
- GPA: 3.90

Activities

1st Place Winner & Presenter, CSUNposium
- Presented the award-winning GoneVishing AI Scam Call Defense System after earning 1st Place at CSUNposium.

Tutor Leader — Los Angeles Pierce College 2023 – 2026
- Led and trained a team of tutors while supporting 20+ students weekly and collaborating with faculty.